

CREDIT RISK ANALYSIS WITH DATA VISUALIZATION USING PYTHON, SQL & POWER BI



PROJECT DESCRIPTION:

Loan analytics based on the bank loan records for the year 2007 through 2015, to review the default rate among loans given to individuals and to get some insights from the data in order to reduce default risk.

I. DATA COLLECTION

- From Kaggle:
- <https://www.kaggle.com/datasets/udaymalviya/bank-loan-data>
- This dataset contains records of loan applicants, with various attributes related to personal demographics, financial status, and loan details. The dataset can be used for predictive modeling, particularly in credit risk assessment and loan default prediction.

2. EXPLORATORY DATA ANALYSIS (EDA) AND DATA PREPARATION VIA PYTHON

- Perform feature engineering and remove inconsistencies.

```
[1]: import numpy as np  
import pandas as pd
```

```
[3]: loan_data_backup = pd.read_csv('data.csv')
```

```
loan_data['emp_length_int']
```

0	10 years
1	< 1 year
2	10 years
3	10 years
4	1 year

```
[33]: loan_data['emp_length_int'] = loan_data['emp_length_int'].str.replace('< 1 year', str(0))  
loan_data['emp_length_int'] = loan_data['emp_length_int'].str.replace('n/a', str(0))  
loan_data['emp_length_int'] = loan_data['emp_length_int'].str.replace(' years', '')  
loan_data['emp_length_int'] = loan_data['emp_length_int'].str.replace(' year', '')
```



```
loan_data['emp_length_int']
```

0	10
1	0
2	10
3	10
4	1

- Change the data types to numeric, datetime or float and create new features

```
[37]: str
```

```
[39]: loan_data['emp_length_int'] = pd.to_numeric(loan_data['emp_length_int'])
```

```
[41]: type(loan_data['emp_length_int'][0])
```

```
[41]: numpy.float64
```

```
[47]: loan_data['earliest_cr_line_date'] = pd.to_datetime(loan_data['earliest_cr_line'])
```

```
[49]: type(loan_data['earliest_cr_line_date'][0])
```

```
[49]: pandas._libs.tslibs.timestamps.Timestamp
```

```
[81]: loan_data['issue_d_date'] = pd.to_datetime(loan_data['issue_d'])  
loan_data['mths_since_issue_d'] = round(pd.to_numeric((pd.to_datetime('2017-12-01') - loan_data['issue_d_date']) / (np.timedelta64(1, 'D'))/30))  
loan_data['mths_since_issue_d'].describe()
```

```
[81]: count      855969.000000  
mean         47.119326  
std          15.196619  
min          35.000000  
25%          35.000000  
50%          47.000000  
75%          55.000000  
max          84.000000
```

- Remove null values

```
: loan_data.isnull().sum()
```

```
: id                0
   member_id        0
   loan_amnt        0
   funded_amnt      0
   funded_amnt_inv  0
```

```
[105]: loan_data['total_rev_hi_lim']=loan_data['total_rev_hi_lim'].fillna(loan_data['funded_amnt'])
```

```
[107]: loan_data['total_rev_hi_lim'].isnull().sum()
```

```
[107]: 0
```

```
[109]: loan_data['annual_inc']=loan_data['annual_inc'].fillna(loan_data['annual_inc'].mean())
```

```
[111]: loan_data['mths_since_earliest_cr_line']=loan_data['mths_since_earliest_cr_line'].fillna(0)
       loan_data['acc_now_delinq']=loan_data['acc_now_delinq'].fillna(0)
       loan_data['total_acc']=loan_data['total_acc'].fillna(0)
       loan_data['pub_rec']=loan_data['pub_rec'].fillna(0)
       loan_data['open_acc']=loan_data['open_acc'].fillna(0)
       loan_data['inq_last_6mths']=loan_data['inq_last_6mths'].fillna(0)
       loan_data['delinq_2yrs']=loan_data['delinq_2yrs'].fillna(0)
       loan_data['emp_length_int']=loan_data['emp_length_int'].fillna(0)
```

3. USING DAX TO CREATE MEASURES

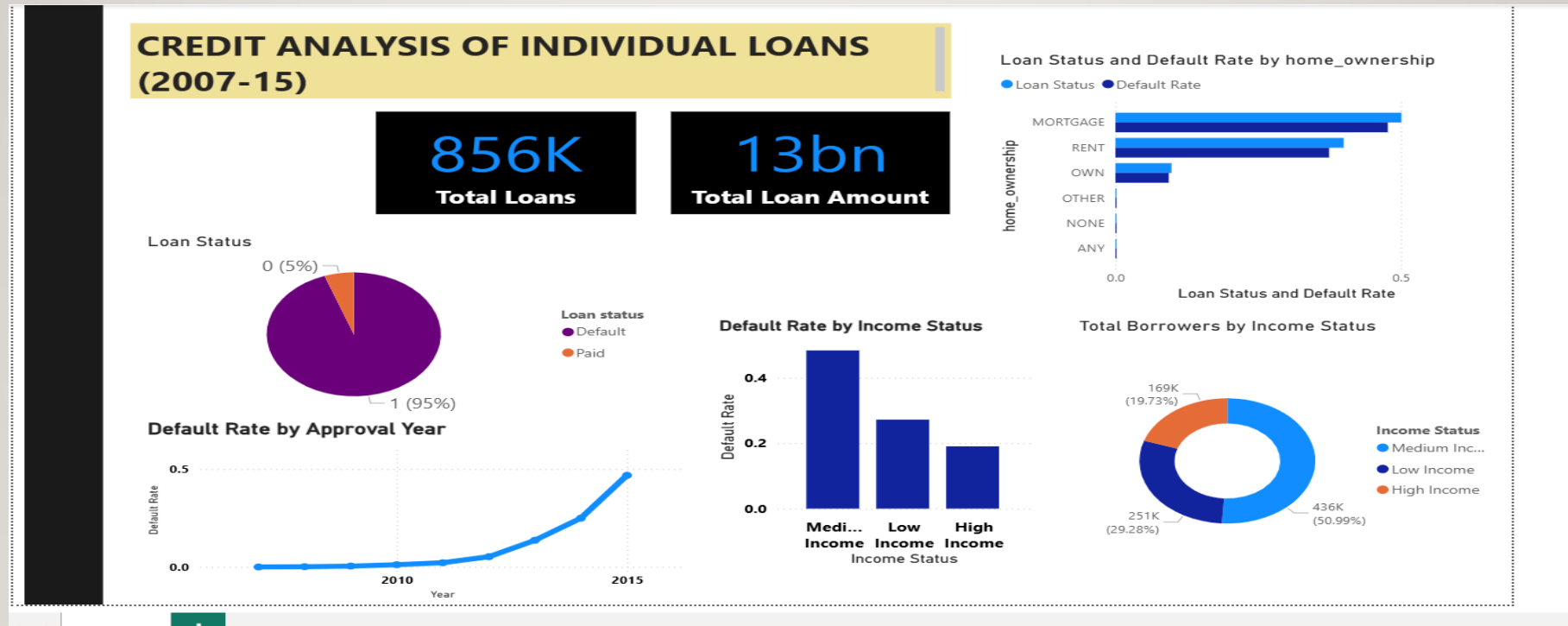
Structure	Formatting
1 Total Loans = COUNT(data[id])	

Structure	Formatting	Properties
1 Loan Status = DIVIDE([Total Loans],[All Loans])		

Structure	Formatting
1 Default Rate = CALCULATE([Loan Status],data[default_ind]=0)	

1 All Loans = CALCULATE([Total Loans],ALL(data))
--

4. DATA ANALYSIS AND VISUALIZATION WITH POWER BI



5. INSIGHTS AND RECOMMENDATIONS

- 1. Default rate is on a rise from 2007-2015. Some measures need to be taken to reduce that.
- 2. 95% of the individuals have defaulted on full loan repayment. Only 5% have paid in full.
- 3. Default rate is found least in individuals that have their own homes.
- 4. Default rate is found to be highest in middle income group.